

HONORS PHYSICS: KINEMATIC RENDEZVOUS CHALLENGE

HS-PS2-1 • SEP-5

Name: _____ Period: _____ Date: _____

UNIT 2: CONSTANT VELOCITY & GRAPHICAL INTERCEPT MODELING • CHALLENGE 1 OF 2

MISSION SCENARIO: EMERGENCY HARBOR AIR-DROP

A marine research ship is traveling straight out to sea away from the harbor dock (dock = $x = 0\text{ m}$). At $t = 0\text{ s}$, the ship is already **300 meters** out from the dock, traveling at a steady speed of **6.0 m/s**.
An urgent medical distress call requires dispatching an autonomous courier drone from the dock ($x = 0\text{ m}$). The drone flies at a steady speed of **18.0 m/s**, but requires a **20-second preflight delay** (boot & rotor spin-up) before taking off at $t = 20\text{ s}$.

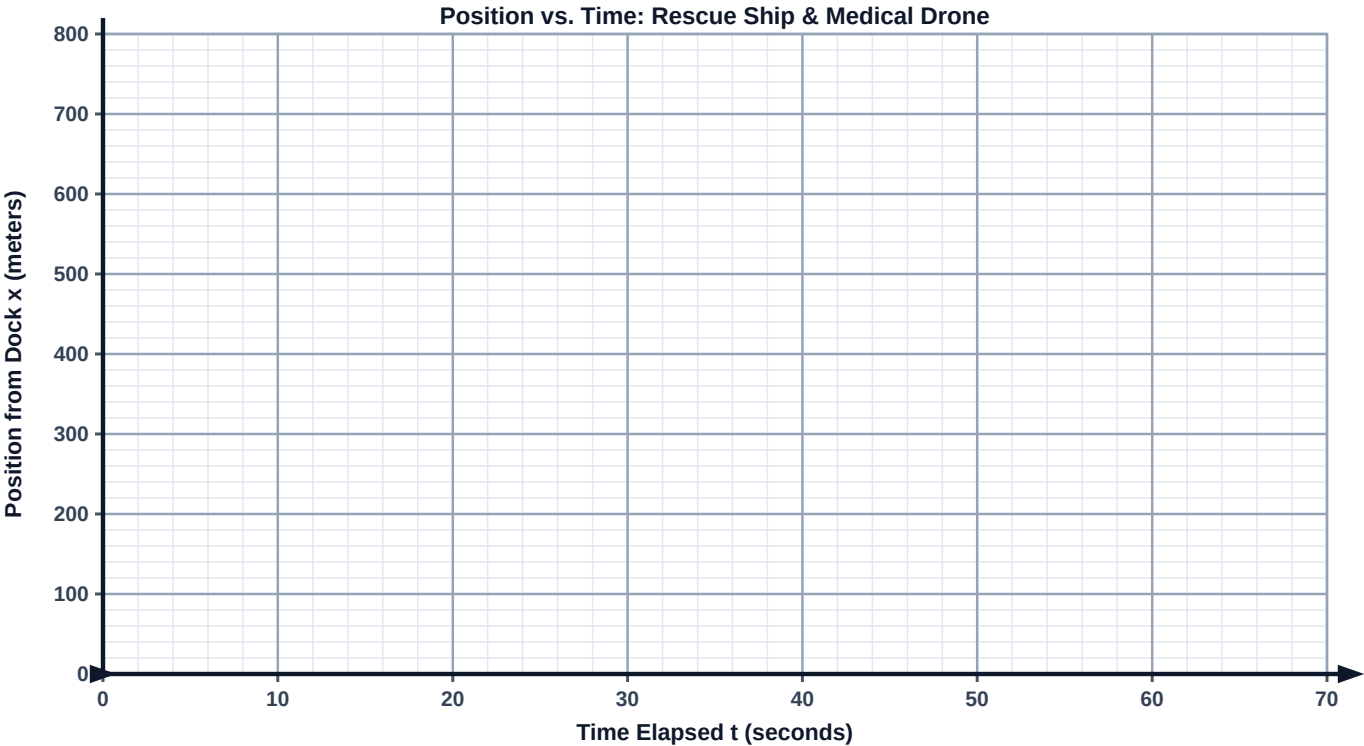
Research Vessel Telemetry

- Initial Position:
 $x_o = 300\text{ m}$
at $t = 0\text{ s}$
- Constant Velocity:
 $v = 6.0\text{ m/s}$
(+60 m every 10 s)

Medical Drone Telemetry

- Initial Position:
 $x_o = 0\text{ m}$
(dock)
- Launch: Holds until
 $t = 20\text{ s}$
, then flies at
 $v = 18.0\text{ m/s}$
(+180 m every 10 s)

Graphing Task: Plot both motions on the grid. Label the Ship line and the Drone line. Circle the intersection point. GRAPH-FIRST ANALYSIS



GRAPHICAL ANALYSIS & MISSION DEBRIEF ANSWER USING YOUR GRAPH ABOVE

- 1. Rendezvous Coordinates (From Graph):**
At what exact time and position do the two lines intersect? **Time (t):** _____ seconds **Position (x):** _____ meters from dock

2. Slope & Speed Verification:
Calculate the slope ($\Delta x / \Delta t$) of each graphed line to verify their speeds: Ship Slope: _____ m/s Drone Slope (after $t = 20\text{ s}$): _____ m/s

3. Actual Drone Flight Duration:
The clock reads your rendezvous time when they meet, but the drone was delayed until $t = 20\text{ s}$. How many seconds was the drone actually flying in the air? _____ seconds

4. Battery Safety Audit (Honors Critical Question):
The drone has enough battery for a maximum round-trip flight distance of **1,200 meters** before complete depletion. Can the drone deliver the cargo and safely return to dock? Explain using numbers:

INTERCEPT

Name: _____

Period: 0

Date: _____

UNIT 2: PIECEWISE 1D MOTION, REST INTERVALS & PURSUIT SLOPES • CHALLENGE 2 OF 2

SURVEILLANCE DOSSIER: DESERT HIGHWAY RECONNAISSANCE

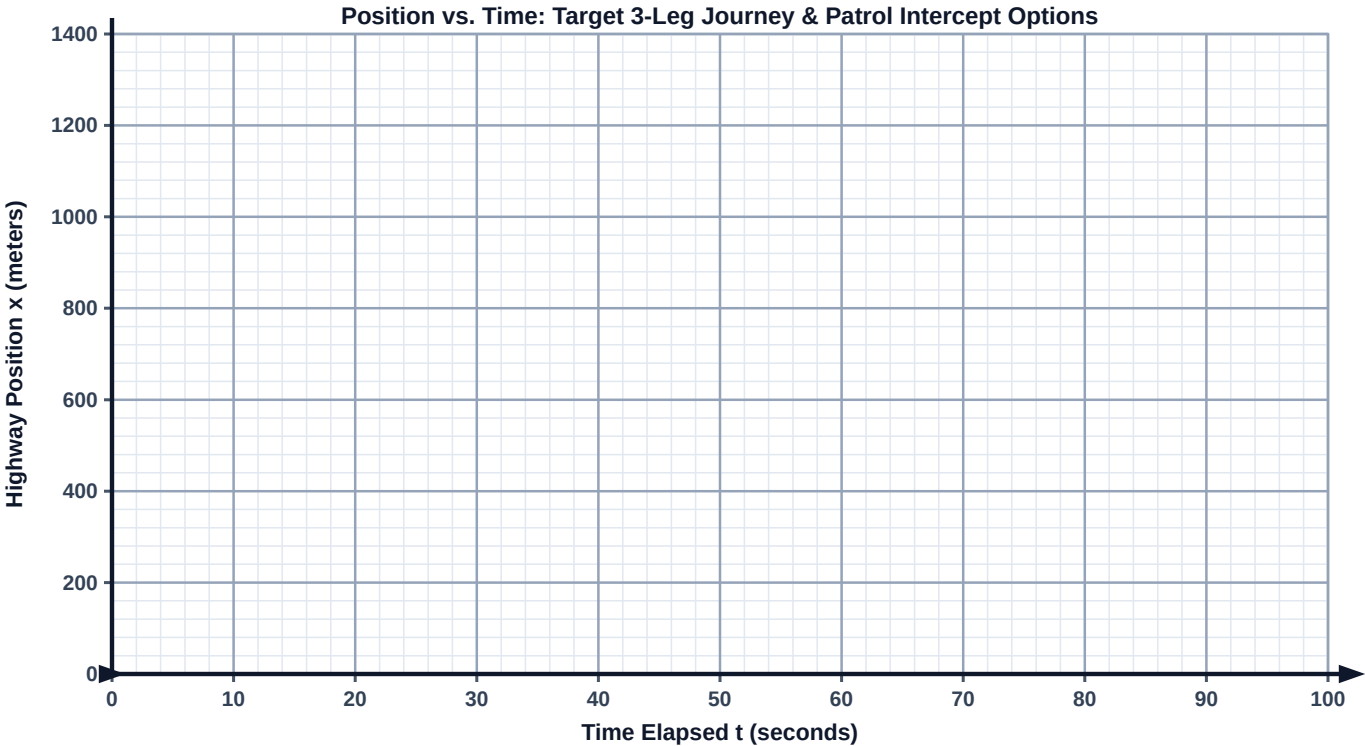
A target vehicle heads east on a straight highway starting from $x = 0\text{ m}$ at $t = 0\text{ s}$ in 3 distinct legs:

- **Leg 1 ($t = 0$ to 20 s):** Drives at 15.0 m/s (covers $15\text{ m/s} \times 20\text{ s} = 300\text{ m}$, reaching $x = 300\text{ m}$ at $t = 20\text{ s}$).
- **Leg 2 ($t = 20$ to 50 s):** Pulls over at a rest stop and remains **completely stopped ($v = 0\text{ m/s}$)** for 30 seconds.
- **Leg 3 ($t = 50$ to 90 s):** Re-enters the highway at 25.0 m/s for 40 s (covers $25\text{ m/s} \times 40\text{ s} = 1,000\text{ m}$, reaching $x = 1,300\text{ m}$ at $t = 90\text{ s}$).

A Highway Patrol Cruiser is at $x = 0\text{ m}$, but is held at the station and **cannot leave until $t = 30\text{ seconds}$** .

 **Graphing Task:** 1) Plot Target's 3-leg line (0 to 90s). 2) Draw Patrol Line A (from $t=30\text{s}$, $x=0$ to catch at rest stop at $t=50\text{s}$). 3) Draw Patrol Line B (to catch at $t=90\text{s}$).

SLOPE AS
VELOCITY



GRAPHICAL ANALYSIS & PURSUIT STRATEGY

ANSWER USING YOUR GRAPH ABOVE

1. Interpreting the Rest Stop (Leg 2):

What does the visual slope of the line represent between $t = 20\text{ s}$ and $t = 50\text{ s}$? How does the graph show the car is stopped?

2. Cruiser Speed Calculations (From Slopes of Lines A and B):

- **Strategy A (Catch at rest stop at $t = 50\text{ s}$):** Cruiser travels 300 m in 20 s (from $t=30$ to 50s). Required speed: _____ m/s
- **Strategy B (Catch at $t = 90\text{ s}$):** Cruiser travels 1,300 m in 60 s (from $t=30$ to 90s). Required speed: _____ m/s

3. Speed Limit Constraint Analysis:

The patrol cruiser has a top safe speed cap of **20.0 m/s (45 mph)**. Which of the two strategies is physically possible? Defend with calculations: